DEPARTMENT OF MATHEMATICS

Final Exam

MTH 240-Calculus II

Last Name (Print):_____ First Name:_____ Student Number: _____

Signature:_____.

Date: June 30, 2015, 6:00 pm

Duration: 90 minutes

Section (circle one)

S. Samiezadeh:	1	2	3
X. Liu:	4	5	6

For Instructor's use only.

Question	Mark
1	
2	
3	
4	
5	
6	
Total	/50

Instructions:

1. This is a closed-book test. Notes, calculators and other AIDS are not permitted.
2. Verify that your test has pages 1-6.
3. (a) Unless otherwise instructed, make sure you include all significant steps in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution.

   (b) Write your solutions in the space provided. If you need more space, use the back of the page, making sure that your solution cannot be confused with any rough work which may be there.
4. Do not separate the sheets.
5. Have your student card available on your desk.

1. [8 marks] Find the radius of convergence and interval of convergence of the series

$$\sum_{n=1}^{\infty} \frac{x^{2n}}{(n^2-2n+3)2^n}$$

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left| \frac{x^{2n+2}}{x^{2n}} \cdot \frac{2^n}{2^{n+1}} \cdot \frac{n^2-2n+3}{n^2+2} \right|$$

$$(n+1)^2 - 2(n+1) + 3 = n^2 + \cancel{2n} + \cancel{1} - \cancel{2n} - \cancel{2} + 3 = n^2 + 2$$

$$\lim_{n\to\infty} \frac{n^2-2n+3}{n^2+2} \overset{H}{=} \lim_{n\to\infty} \frac{2n-2}{2n} = 1$$

$$= \lim_{n\to\infty} \left| x^2 \cdot \frac{1}{2} \cdot 1 \right|$$

$$= \frac{x^2}{2} < 1$$

$$x^2 < 2 \longrightarrow -\sqrt{2} \overset{?}{<} x \overset{?}{<} \sqrt{2} \longrightarrow R = \sqrt{2}$$

@ $x = -\sqrt{2} \Rightarrow \sum_{n=1}^{\infty} \frac{2^n}{(n^2-2n+3)2^n} = \underbrace{\sum_{n=1}^{\infty} \frac{1}{n^2-2n+3}}$

$\sum a_n \,/\, \sum b_n = \underbrace{\sum \frac{1}{n^2}}$

p-series (p=2) Conv.

$$\lim_{n\to\infty} \frac{\frac{1}{n^2-2n+3}}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n^2}{n^2-2n+3} = 1 > 0$$

$\sum a_n$ is conv. by LCT.

@ $x = \sqrt{2} \longrightarrow$ The same series $\longrightarrow$ conv.

Interval of Conv: $[-\sqrt{2}, \sqrt{2}]$ or $-\sqrt{2} \leq x \leq \sqrt{2}$

2. [8 marks] Find a power series representation for $f(x) = \ln \frac{1+x}{1-x}$ and determine the interval of convergence.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad \to \underbrace{\ln(1+x)}_{\text{II}} - \underbrace{\ln(1-x)}_{\text{I}}$$

$$\int \frac{1}{1-x}\,dx = \int \sum_{n=0}^{\infty} x^n\,dx \to -\ln|1-x| = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} + \cancel{C}_{0}$$

$$\boxed{-\ln(1-x) = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}} \quad \text{I}$$

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n$$

$$\int \frac{1}{1+x}\,dx = \int \sum_{n=0}^{\infty} (-1)^n x^n \Rightarrow \boxed{\ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}} + \cancel{C}_{0}$$

II

$$= \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} + \frac{(-1)^n x^{n+1}}{n+1} = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}\left[1 + (-1)^n\right]$$

$\to 0$ odd n

$\to 2$ even n

$$= 2 \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}$$

3. [8 marks] Find the Maclaurin series for the function $f(x) = \frac{x}{\sqrt{4+x^2}}$ and its radius of convergence.

   (Hint: Use the Binomial Series)

4. [5+5 marks] a) Approximate $f(x) = \cos 2x$ by a Taylor polynomial $T_4(x)$ with degree 4 at $\frac{\pi}{2}$

$$T_4(x) = \sum_{n=0}^{4} \frac{f^{(n)}(\frac{\pi}{2})}{n!}\left(x - \frac{\pi}{2}\right)^n$$

$$= -1 + \frac{4}{2!}\left(x - \frac{\pi}{2}\right)^2 - \frac{16}{4!}\left(x - \frac{\pi}{2}\right)^4$$

$$= -1 + 2\left(x - \frac{\pi}{2}\right)^2 - \frac{2}{3}\left(x - \frac{\pi}{2}\right)^4$$

n	$f^{(n)}(x)$	$f^{(n)}(\frac{\pi}{2})$
0	$\cos 2x$	-1
1	$-2\sin 2x$	0
2	$-4\cos 2x$	4
3	$8\sin 2x$	0
4	$16\cos 2x$	-16
$\rightarrow$ 5	$-32\sin 2x$	0

$$\frac{\pi}{2} - 1 \leq x \leq \frac{\pi}{2} + 1$$

$$M = f^{(5)}(x)$$

b) Use Taylor's Inequality to estimate the accuracy of the approximation $f(x) \approx T_4(x)$ when $\frac{\pi}{2} - 1 \leq x \leq \frac{\pi}{2} + 1$.

$$|R_n(x)| \leq \frac{M}{(n+1)!}\underbrace{\left|x - \frac{\pi}{2}\right|^{n+1}}_{=1}$$

$$f^{(5)}(x) = -32\sin 2x$$

M = max of $(-32\sin 2x)$ on $\frac{\pi}{2} - 1 \leq x \leq \frac{\pi}{2} + 1$

$= 32$

$$\text{error} \leq \frac{32}{5!}(1) = \frac{32}{5 \times 4 \times 3 \times 2} = \frac{32}{120}$$

5. [8 marks] Use the Chain Rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

$$a = e^r \cos\theta, r = st, \theta = \sqrt{s^2 + t^2}$$

6. [8 marks] Find the absolute maximum and minimum values of $f(x, y) = 2x^3 + y^4$ on the set $D = \{(x, y)|\ x^2 + y^2 \leq 1\}$.